Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To Eliminate concern of console to IP gap uneven in Scorpio model"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio TCF.

	Plan	Actual
Start Date	**10.10.18**	**10.10.18**
Completion Date	**30.12.19**	**30.12.19**

Current Status: **CLOSED**

Cross Functional Team

Sr. No	Description		Name	Designation	Dept.	Phone
1	Team Leader	Amit V.Tile	DY. Manager	Scorpio TCF Prod.	9923282427	Tile.amit@mahindra.com
2	Mentor	Shaikh Taj	Sr. Manager	Scorpio TCF Prod	9850980299	Shaikh.Taj@mahindra.com
3	Member 1	Vikrant Ghule	DY. Manager	Scorpio TCF Prod.	9923282427	Ghule.viktrant@mahindra.com
4	Member 2	Mandar Joshi	Manager	Scorpio TCF Prod.	9923282427	Joshi.mandar@mahindra.com
5	Member 3	Nilesh deore	Ass. Manager	Scorpio TCF CAI	8888821705	Deore.nilesh@mahindra.com
6	Member 4	Santosh Jadhav	DY. Manager	Scorpio TCF MQA	9923282427	Jadhav.santosh8@mahindra.com

Step1. Select Subject

Brief description of Problem:-

"To Eliminate concern of IP to console Gap in Scorpio model"

What : Excess gap between IP top console .

Where : At CAI .

When : Since last Nov-18.

Who : By associate at console stage, IP fitment stage .

WHY : High rework, STR loss, Repeat issue at CAI .

HOW : Avg. 21 cases/month reported at CAI and 6-7 cases at RFI /GNOVAC .

IP to console gap

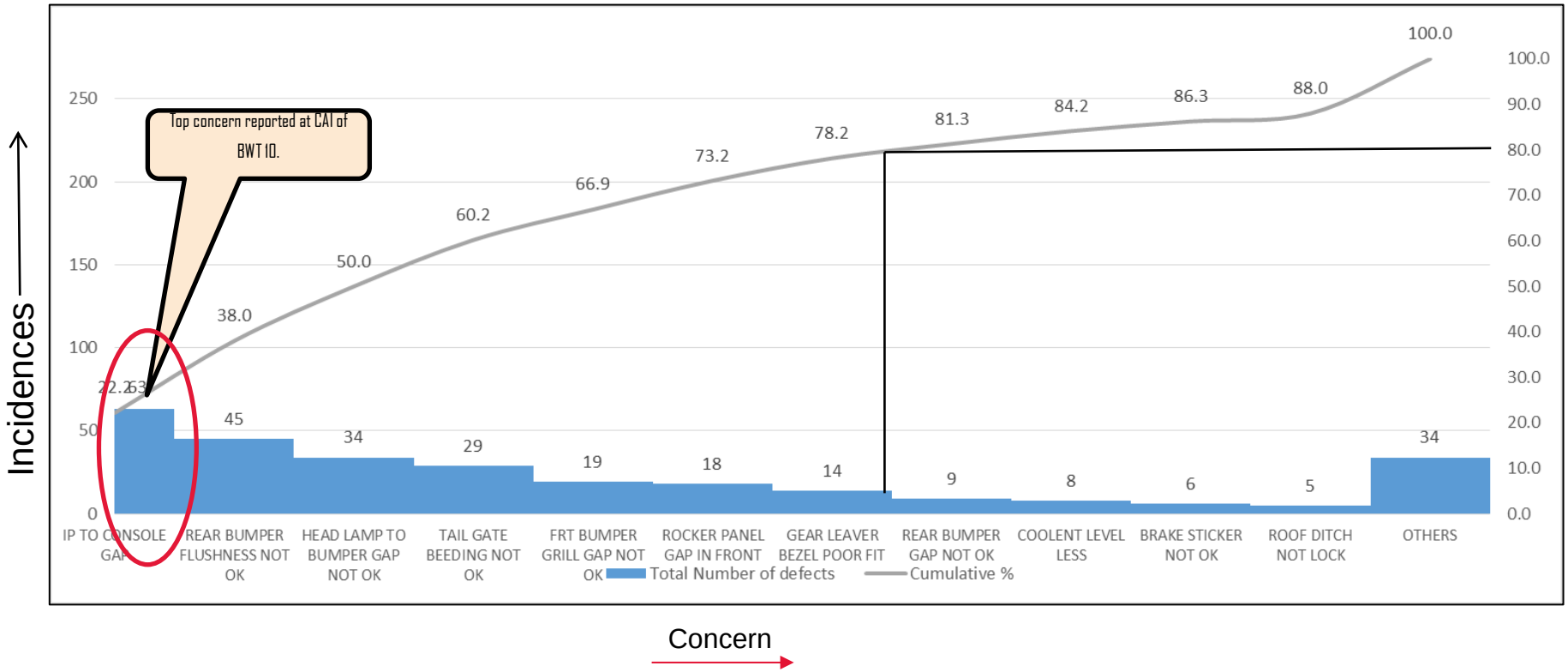
2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice

Pareto analysis for BWT-10 concerns reported in CAI



Inference:
Console to IP gap not ok concern reported at CAI is top concern of BWT 10 process concern. Above data shown for 10 concern booked from NOV-18 to JAN-19

2. Justify the choice

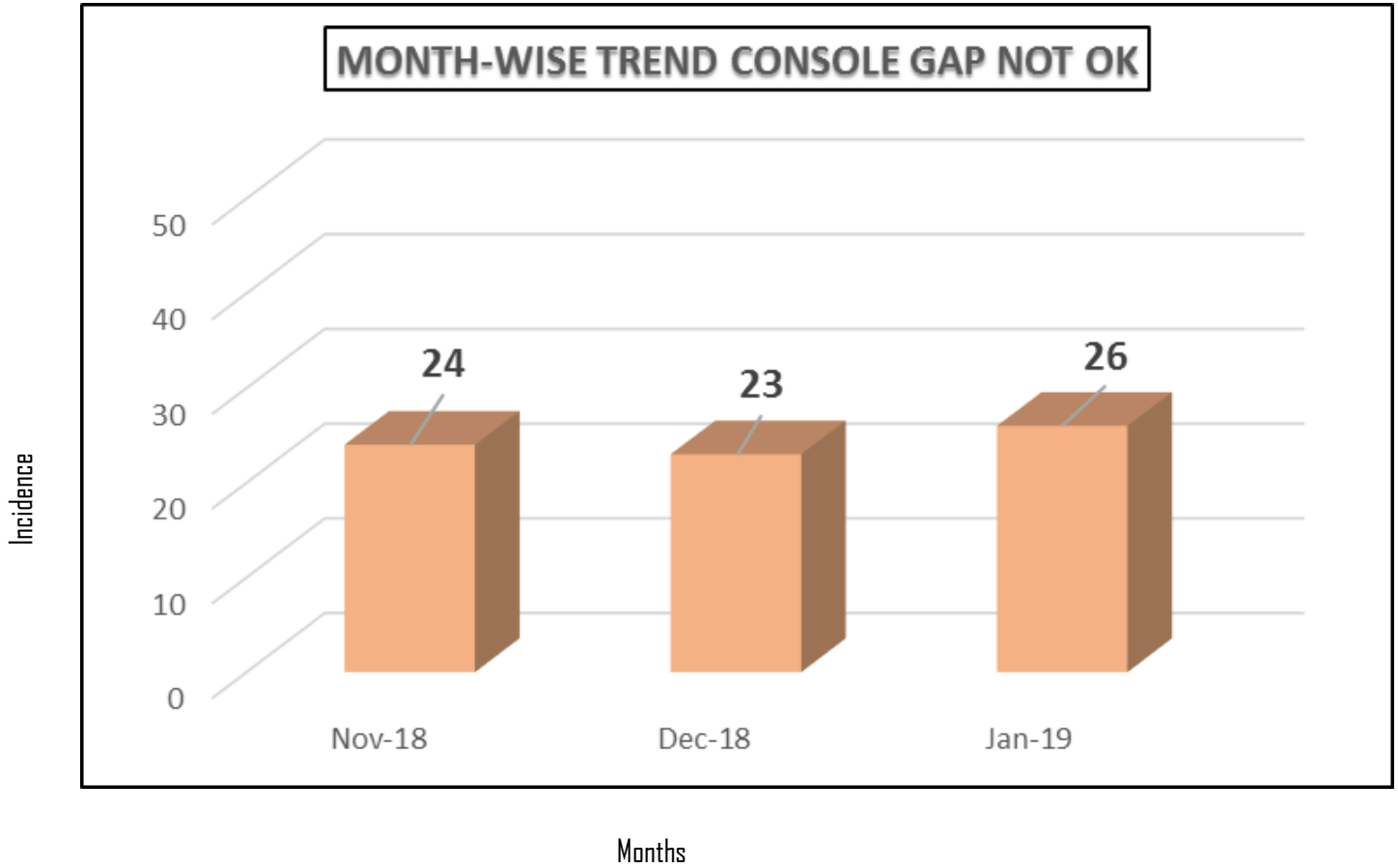
Severity & Consequences of Concern :-

- Avg. 21 Cases reported per month, for rework, 6-7 V2 concern reported in Flash aves audit. Avg concern per day 3-5 reported at buyoff.
- Customer Dissatisfaction due to Bad aesthetics.
- Incorrect appearance when customer enters or seat inside veh.
- Rework Difficult
- Rework cost more.
- Rework time 30 min (approx.) for each vehicle by 1 operator.
- Rework cost is Rs.87097/- per year

COPQ (Cost of Poor Quality) Calculation

Description	Units	
Rewok Time	Min	30
Avg Concerns/Day	Nos	3
Total Required Rework Time per day	Min	90
Effective Working time	MIn	465
Operator Require Per Day	Nos	0.19
Approximate Cost Per Day	Rs	1500
Operator Cost/ Concern	Rs	290.32
Avg no of working days	Days	25
Operator Cost Per Month	Rs	7258.06
Operator Cost /Year	Rs	87,097

2. Justify the choice

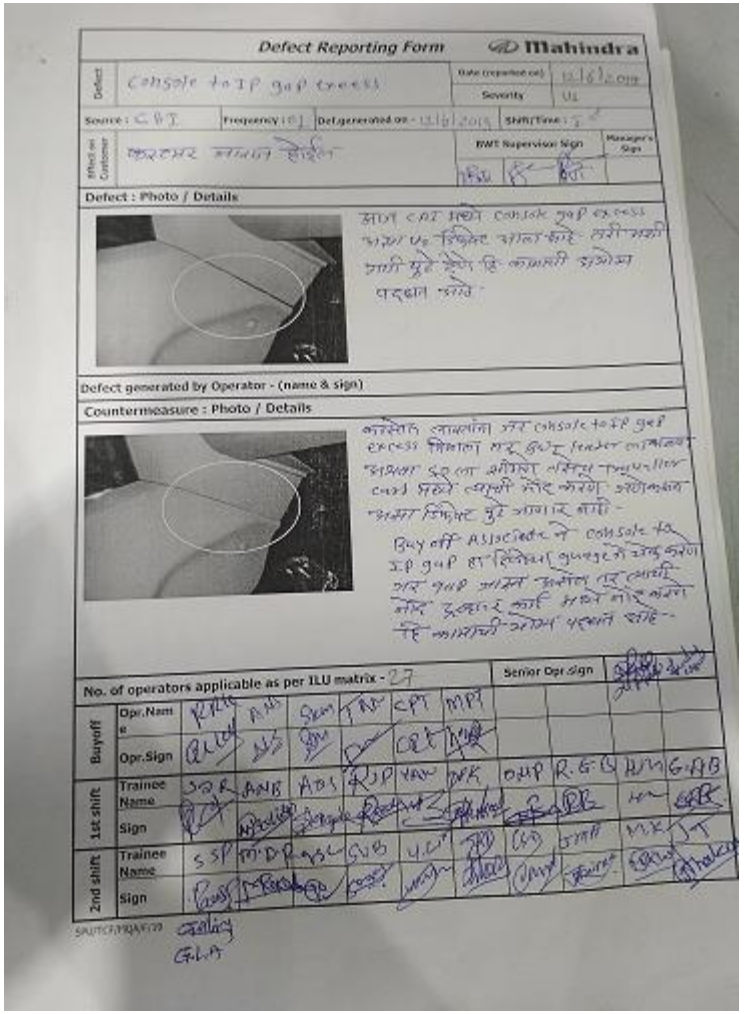


Inference : Average 80 Nos concerns reported per month from NOV -18 to JAN-19

2.b Immediate Containment Action

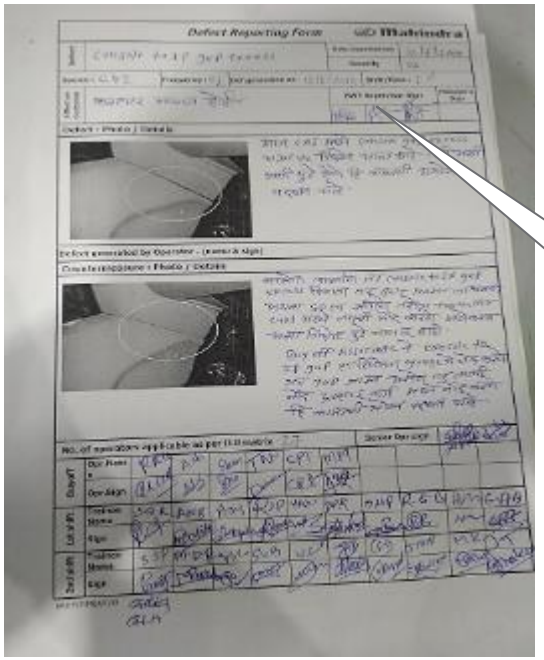
- Concern communicated to stage associates through DRF.

- Training for stage associates, explaining main step and key points.
- Emphasis on part handling and Tool handling while doing work.
- 100% part to be check at in process during console fitment.
- DRF given to stage associates, explaining key points and focus on Gap during fitment.



2.b Immediate Containment Action

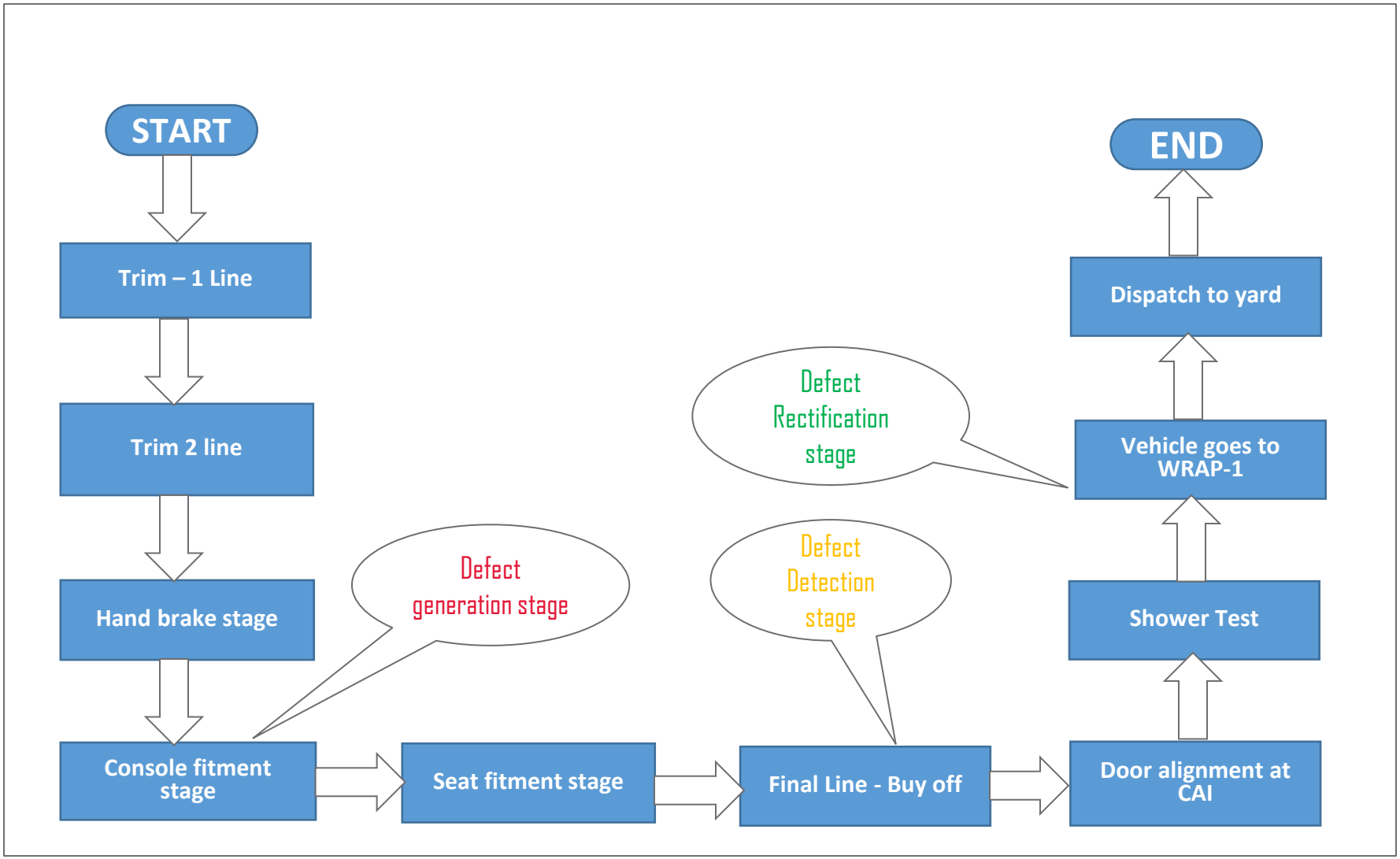
- ❑ Concern communicated to stage associates through DRF.
- ❑ Checkpoint added at Buy-off to arrest defect passing on to next stage.
- ❑ Gauge provided for checking console to IP gap



Gauge provided for checking gap between console to IP gap

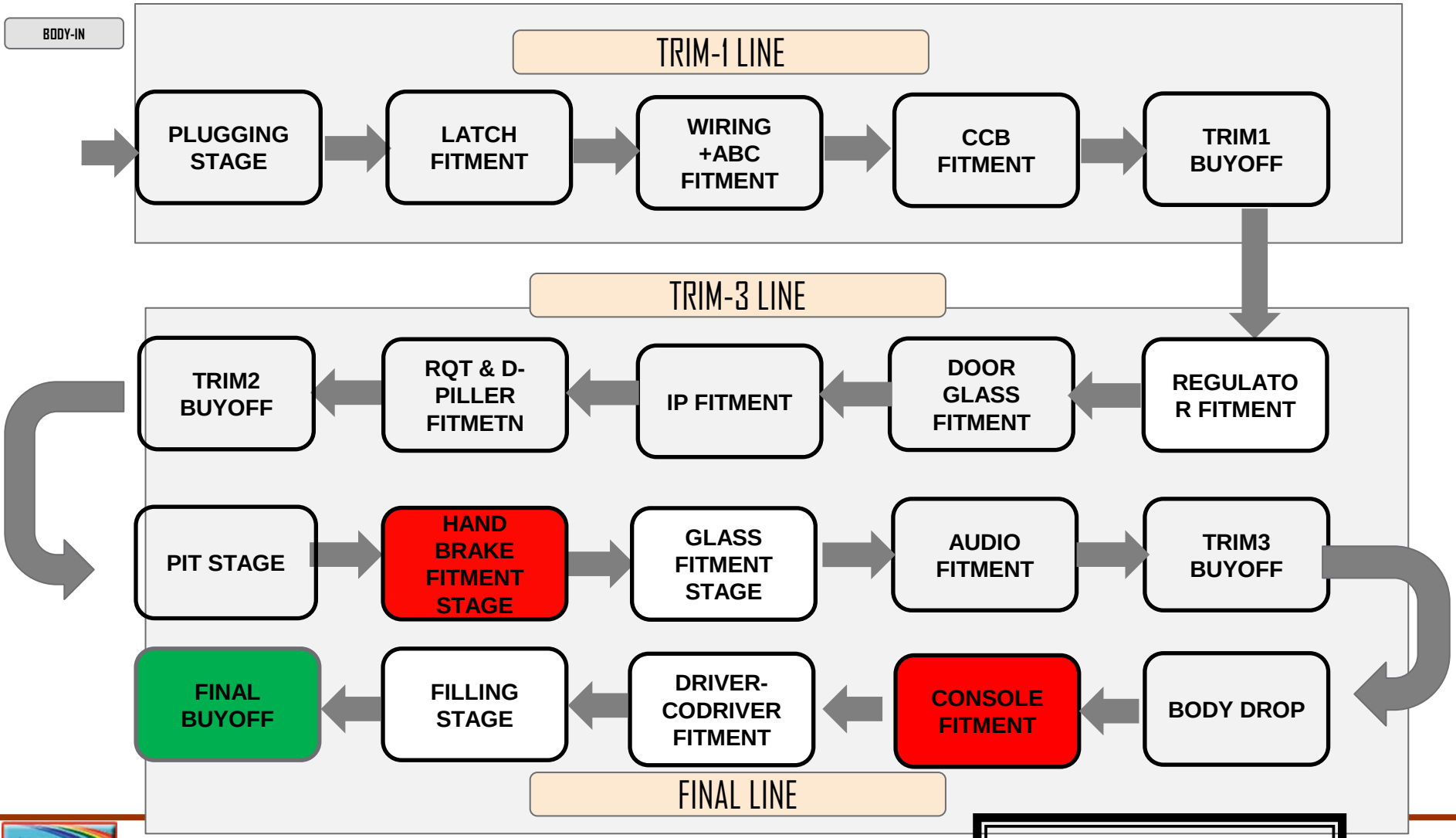
Communication through DRF

Step 3. Understanding Current Situation



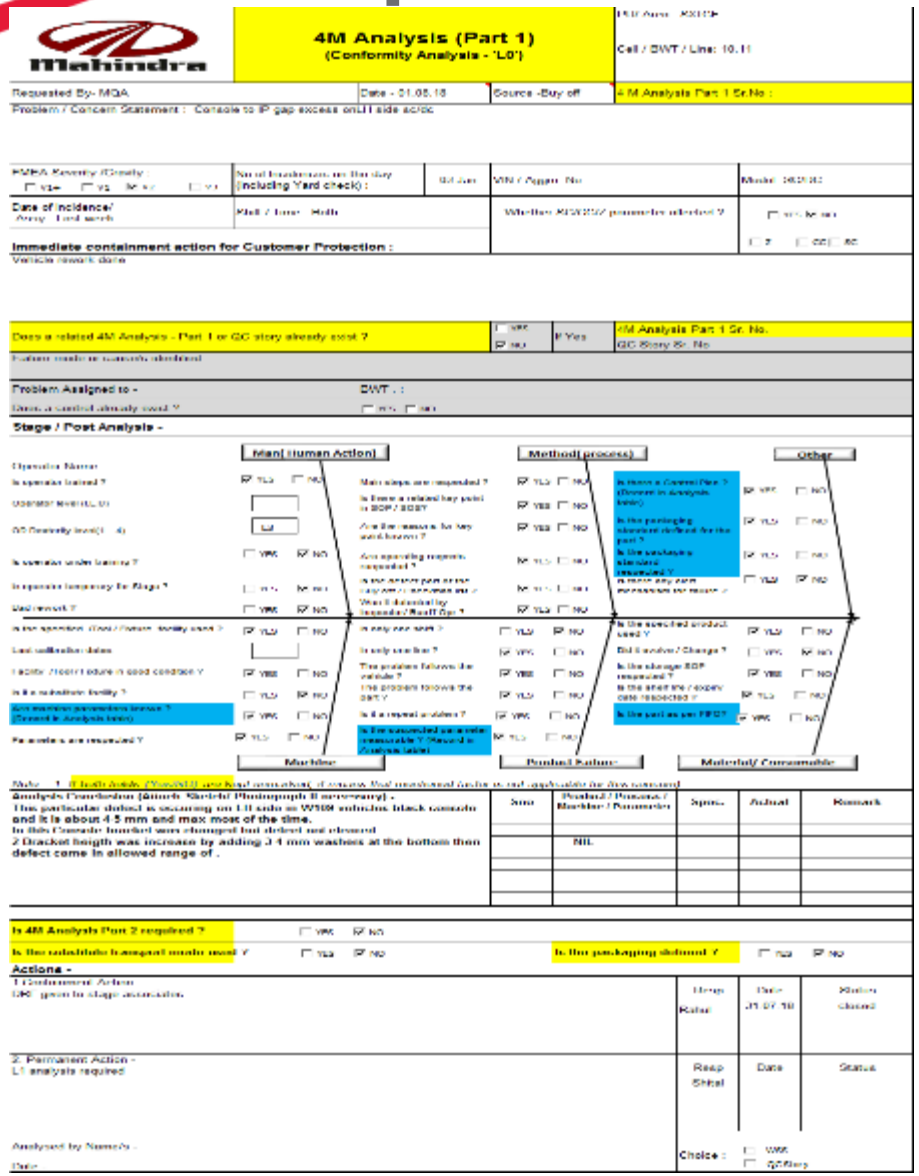
Step 3. Understanding Current Situation

Observation of process



Defect generation stage
Defect identification stage

Step 3. Understanding Current Situation



4M analysis Good and Not so good observations	
MAN	Operator dext. Level L3
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows Part
Material	Handling and shelf life of part adhered

Step 3. Understanding Current Situation

- Issue observed in both Scorpio domestic and export Model.

- Issue reported in both the shifts.

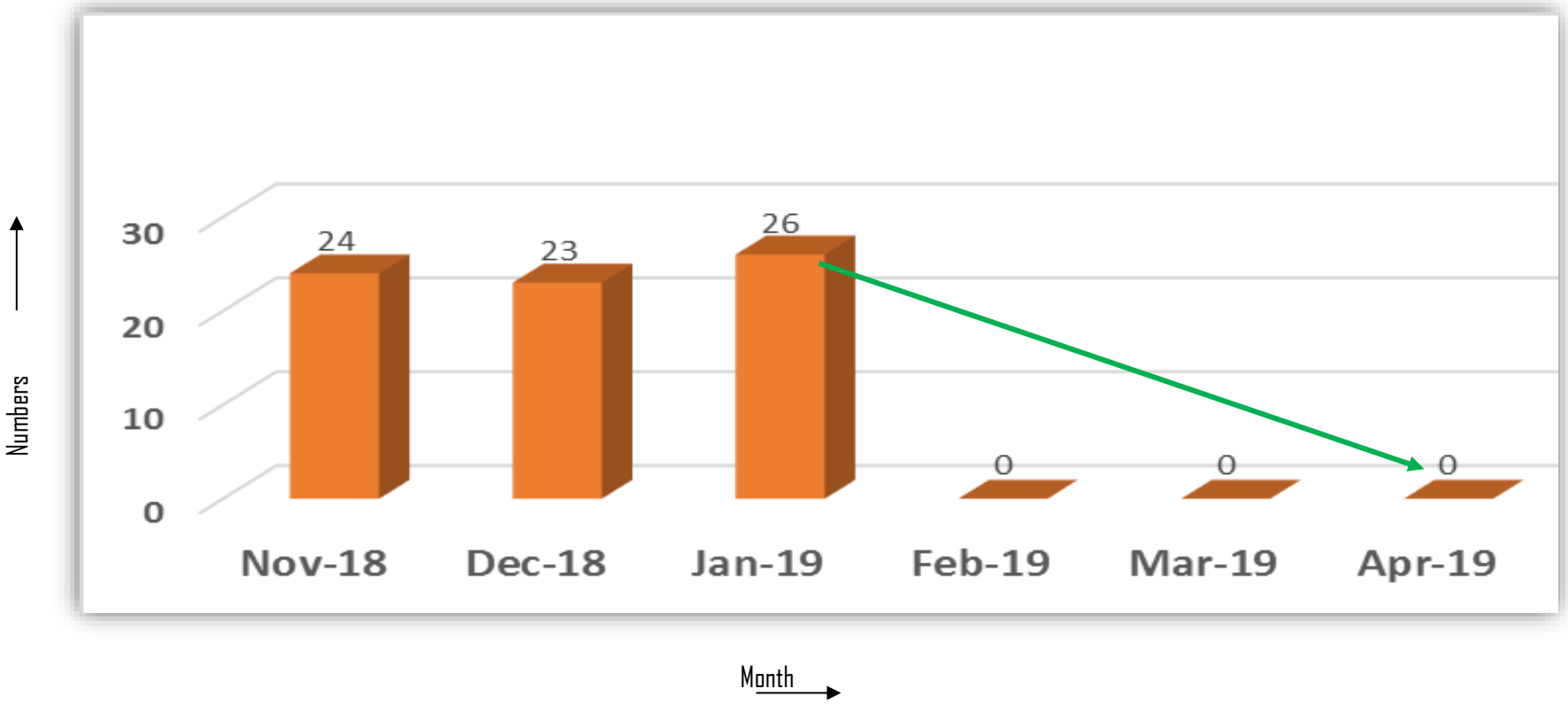
- Repair time is high, around 30 minutes per vehicle.(To maintain gap console need to take out from vehicle and adjust the center console mtg. bracket.)

- Level 3 Associates are deployed for console fitment .

- While during console fitment excess gap observe between IP to console .

3. Understand the current situation

Present Level:	**Avg. 21 Incidences are being reported at CAI /month.**
Target Level:	**Eliminate concern Gap between IP to console by end of APR-19.**



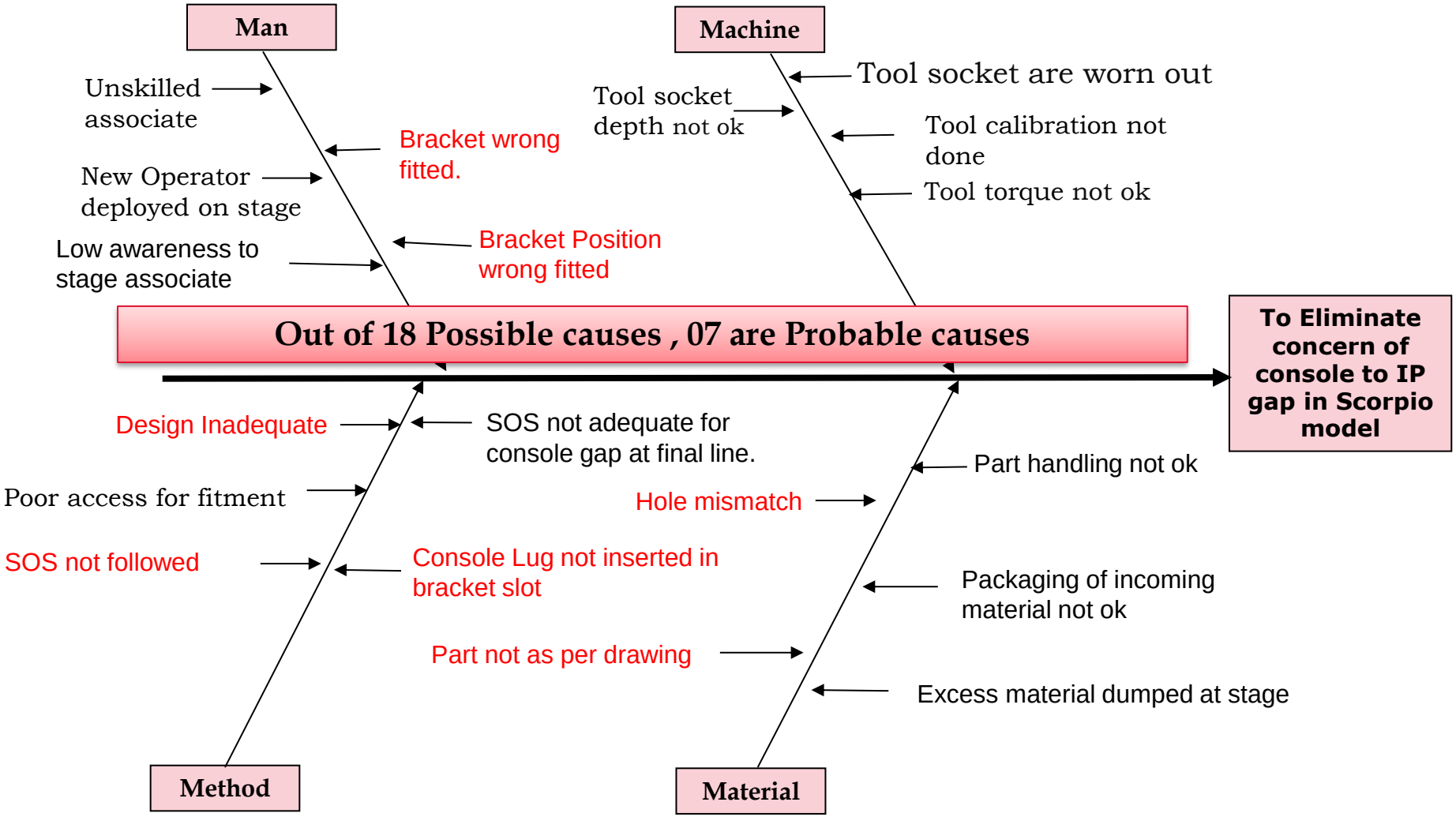
5. Analysis

List of Possible causes after brainstorming among team members

Sr. no	Possible Causes
1	Unskilled associate
2	New Operator deployed on stage
3	Low awareness to stage associate
4	Tool socket depth not ok
5	Tool socket wear out
6	Tool calibration not done
7	Tool torque not ok
8	More gap between console and bracket
9	Poor access for fitment
10	SOS not followed
11	Hole mismatch
12	Bracket wrong fitted
13	SOS not adequate
14	Part not as per drawing
15	Part hole mismatch
16	Part handling not ok
17	Packaging of incoming material not ok
18	Excess material dumped at stage.
19	Wrong position fitted

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr. No.	Possible Cause	Amit	Taj	Santosh	Mandar	Nilesh	Chandu	Vikrant	Total
1	Unskilled Associate	3	1	3	3	3	3	3	19
2	New associate deploy on stage	1	3	3	3	3	3	3	19
3	Low awareness to stage associate	1	1	1	3	1	1	3	11
4	Tool socket depth not ok	1	1	3	1	1	3	1	11
5	Tool socket wear out	1	1	1	3	1	3	1	11
6	**Bracket position not ok**	9	3	9	9	3	3	3	39
7	Tool torque not ok	3	1	1	1	1	1	3	11
8	**Design inadequate**	9	3	9	9	9	3	1	43
9	Poor access for fitment	3	3	9	1	1	1	1	19
10	**SOS not adequate**	9	9	3	3	3	9	9	45
11	**Hole missmatch bkt to Console**	9	9	9	3	3	3	3	39
12	**Bracket wrong fitted**	9	9	9	3	3	9	3	45
13	SOS not followed by associate.	1	3	3	3	1	1	9	21
14	**Console lug not inserted into bkt slot**	9	9	9	3	3	9	3	45
15	**Part not as per drawing**	9	3	9	9	3	3	3	39
16	Part handling not ok	9	9	3	3	1	3	3	31
17	Packaging of incoming material not ok	1	1	1	1	1	1	1	7
18	Exsess material dumped at stage.	1	1	1	3	1	1	3	11

Scale: 0=None, 1=Low, 3=Moderate, 9=Strong
Threshold Limit = 39

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

8	Design Inadequate
10	SOS not adequate
11	Hole missmatch bkt to Console
12	Bracket wrong fitted
14	Console lug not inserted into bkt slot
15	Part not as per drawing
17	Bracket position not ok

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr. No	Not considered Causes	Reasons
1	Unskilled Associate.	No new associate deployed n stage, Dexterity level confirmed. Ensure model training completion
2	New Associate deployed on stage.	All trained associate deployed on stage
3	Low awareness to stage associate	Found ok
4	Tool socket not ok	Tool socket found ok
5	Tool socket wear out	Tool socket check and found ok
6	Tool calibration not done	Tool calibration stciker check and found ok
7	Tool torque not ok	Tool torque check and found ok as per specification
8	Poor access for fitment	No issue observed-30veh checked
9	SOS not adequate	SOS are followed
10	Part handling not ok	Visually checked Found ok
11	Packaging of incoming material not ok	Incoming part check and found ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Design inadequate	Vehicle wise data monitoring	On observation it was observe that there is gap between console and mounting bracket	**Hypothesis valid**

Concern	Console are resting on bracket ok	Console are not resting on bracket
Vehicle Checked	|||||||||||||||||||||||||||||||						

- **Total 30 nos of vehicle checked for console resting on console mounting bracket. Out 30 6 vehicle found console not resting completely on mounting bracket. Gap has been observed between console and bracket.**

After Data Analysis we have found issue is only observed in mhawk model and not in LCCR model.

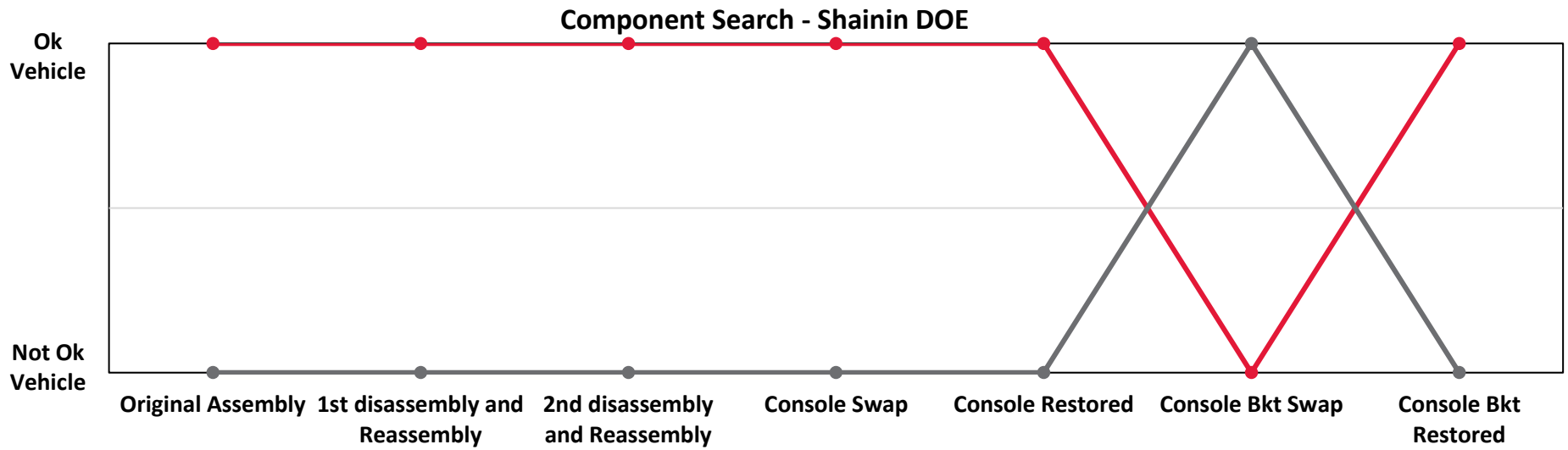
Gap observe between console and console bracket

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Design inadequate	Vehicle wise data monitoring	On observation it was observe that there is gap between console and mounting bracket	**Hypothesis valid**

CFT decided to do DOE analysis for Ok & Not Ok vehicles



Defect followed with Console Bkt.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	SOS inadequate for console fitment	Process Verification along with CFT	During observation it is found that if console duct not inserted into IP Blower then gap gets created between console and IP .	**Hypothesis Valid**

Console not inserted properly and fitted on duct due to this gap observe.

Concern	Gap found ok as per sos	Gap found not ok as per sos
Vehicle Checked	|||||||||||				

- **Associate followed the sequence of operation but did not maintain the gap between IP and console**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Hole mismatch	Visual.	Total 25 vehicle checked and it is found that hole was align and found concentric at time of fitment	**Hypothesis Invalid**

Vehicle check in both shift and found that bracket slot and console hole was concentric with each other and lug was properly inserted into it.

Concern	Lug inserted into bkt hole	Lug not inserted into bkt hole
Vehicle Checked	|||||||||||||||||||||||||||	

- **Total 25 nos of vehicle checked and found ok**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Bracket wrong fitted.	Visual	Total 20 nos of vehicle checked in both shift and it is observe bracket can be wrong fitted as per model .	**Hypothesis valid**

Concern	Console Bkt as per model	Console Bkt not as per model
Vehicle Checked	|||||||||||||||||||||	

Bracket fitted for S3 ,

Bracket fitted for higher variant such as S4,S5,S7, S9 , S11,

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Console lug not inserted into bracket slot	Visual inspection	25 nos of vehicle checked found it is found that lug get inserted into bracket hole	**Hypothesis Invalid**

Console lug has to insert into console bracket slot such that it will match with each other. During observation it is found ok.

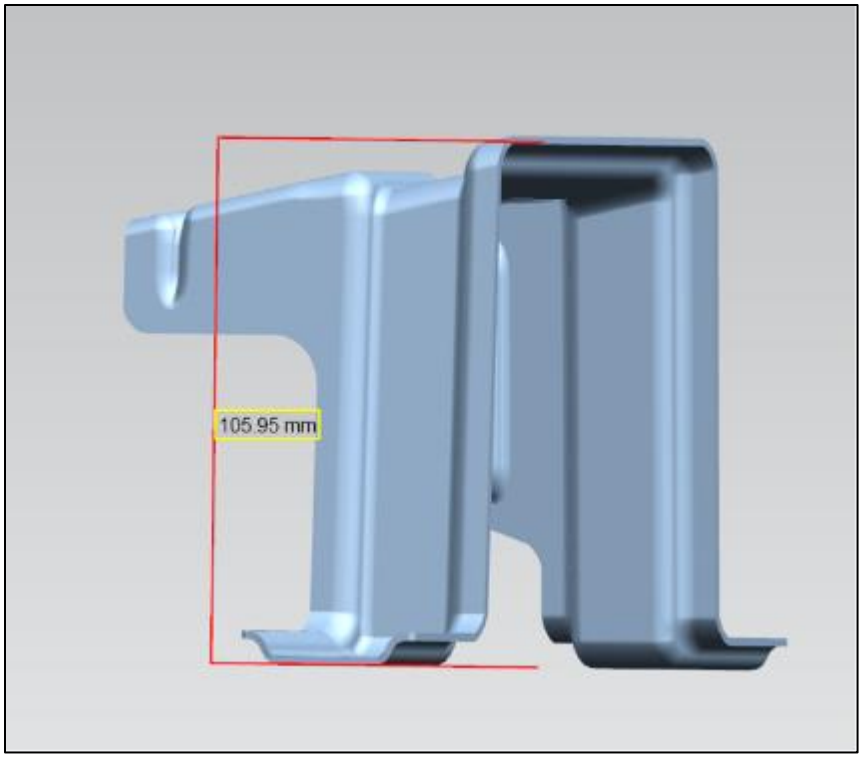
Concern	Lug inserted into bkt hole	Lug not inserted into bkt hole
Vehicle Checked	~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~	

- **Associate does respect the main step and key point for console fitment.**
- **Total 20 vehicle process observed with 6 different associate in both shift, associate follow the key point and the reason for key point. Hence hypothesis are Invalid.**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
6	Part not as per drawing	Drawing verification	Total 15nos of part check and it is found that part is as per drawing.	**Hypothesis Invalid**



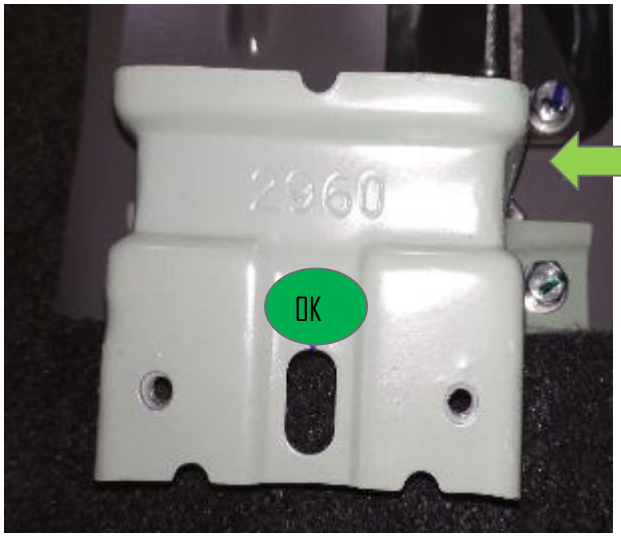
Console bkt height	
Sr.No.	Height In mm
1	105.85
2	105.91
3	106.08
4	105.91
5	105.82
6	105.94
7	105.91
8	105.95
9	105.92
10	105.89
11	105.92
12	105.95
13	105.94
14	106.02
15	105.91

- **Total 15 no's height of bracket checked by Vernier scale.**
- **All parts are found within tolerance limit.**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
7	Bracket position wrong fitted	Visual	Total 15nos of part check and it is found that bracket can be fitted wrong .But console fitment is not happened.	**Hypothesis Invalid**



Correct Bracket position fitment

Bracket position can be fitted wrong

- **Total 15 nos of vehicle checked for console bracket position wrong in different model but console not get properly rested on bracket hence console not fitted.**

Concern	Console not fitted with bkt position wrong	Console fitted with bkt position wrong
Vehicle Checked	|||||	

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
1	Gap between console and mounting bracket	Visual check	Hypothesis valid
2	SOS inadequate for console fitment	Process audit Go through process	Hypothesis valid
3	Hole mismatch	Visual.	Hypothesis Invalid
4	Bracket wrong fitted.	Visual	Hypothesis valid
5	Console lug not inserted into bracket slot	Visual inspection	Hypothesis Invalid
6	Part not as per drawing	Drawing verification	Hypothesis Invalid
7	Bracket position wrong fitted	Visual	Hypothesis Invalid

WHY - WHY Analysis

5.3 Identify the root causes:

Sr. No	Probable cause	WHY?	WHY?	WHY?	Action
1	Design Inadequate	Issue only observed in mHawk vehicles	Console bkt height is different	Due to wiring layout for 12V Socket it is made different but is creating issue	Bracket design to be change
2	SOS inadequate for console fitment	Console front side edge not matching correctly	No specific check or control point given to associates	This AC duct alignment failure mode was not known	Correct SOS and give awareness
3	Bracket wrong fitted.	Difficult to identify bracket as per model though color is different	No significant difference in parts	--	Part to be commonise

6. Implement the corrective measures

Possible Remedies Identified

Sr. No.	Probable Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Due to wiring layout for 12V Socket it is made different but is creating issue	Bracket design to be change	Low	Low	Yes	Low	Accepted
2	This AC duct alignment failure mode was not known	Correct SOS and give awareness	Low	Low	Yes	High	Accepted
3	No significant difference in parts	Part to be commonise	High	Low	Yes	High	Accepted

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1	Due to wiring layout for 12V Socket it is made different but is creating issue	W110 S3 model Bkt to be made common	29.6.2019	Santosh	Closed

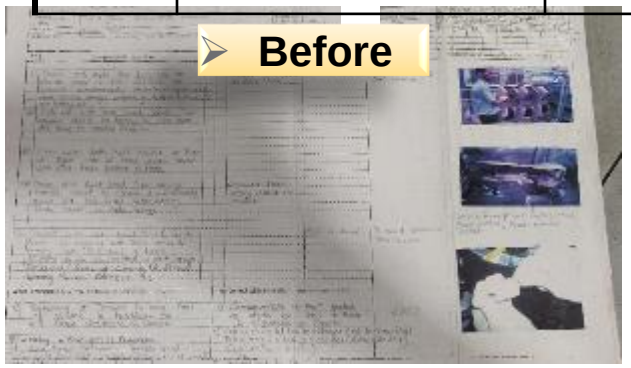
Height of bracket to be increased by 8 mm

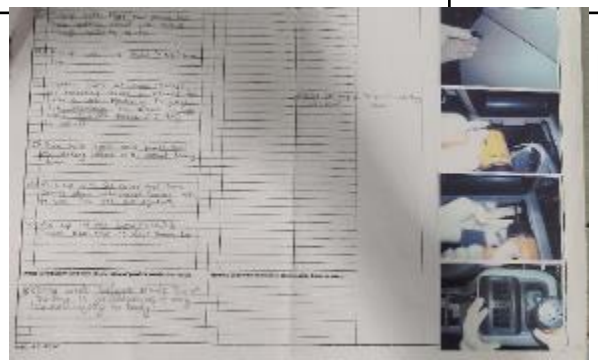
6. Implement the corrective measures

6.1 Actions Decided

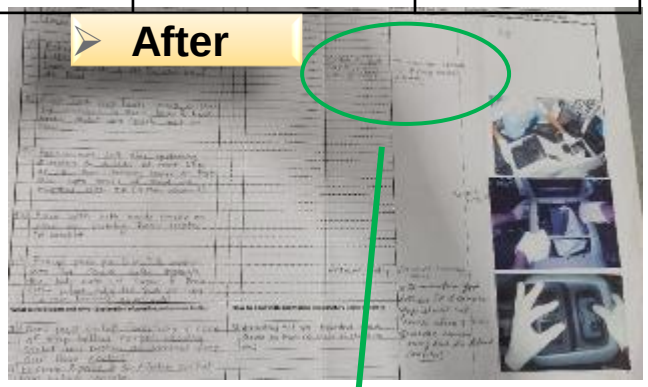
Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
2.	This AC duct alignment failure mode was not known	sos corrected for console ac duct matched with the IP duct.	9.7.2019	Amit tile	Closed

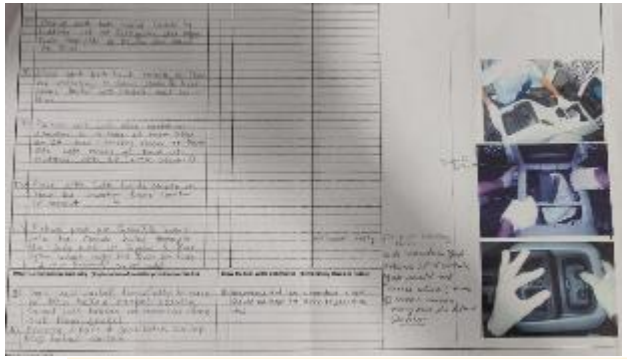
➢ **Before**

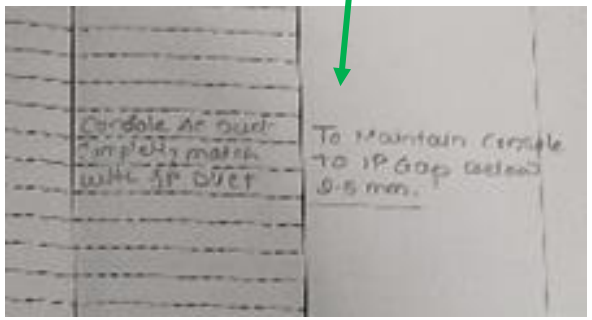




➢ **After**







➢ **Point no. 7 modified for key point, IP and console AC duct completely match with each other**

6. Implement the corrective measures

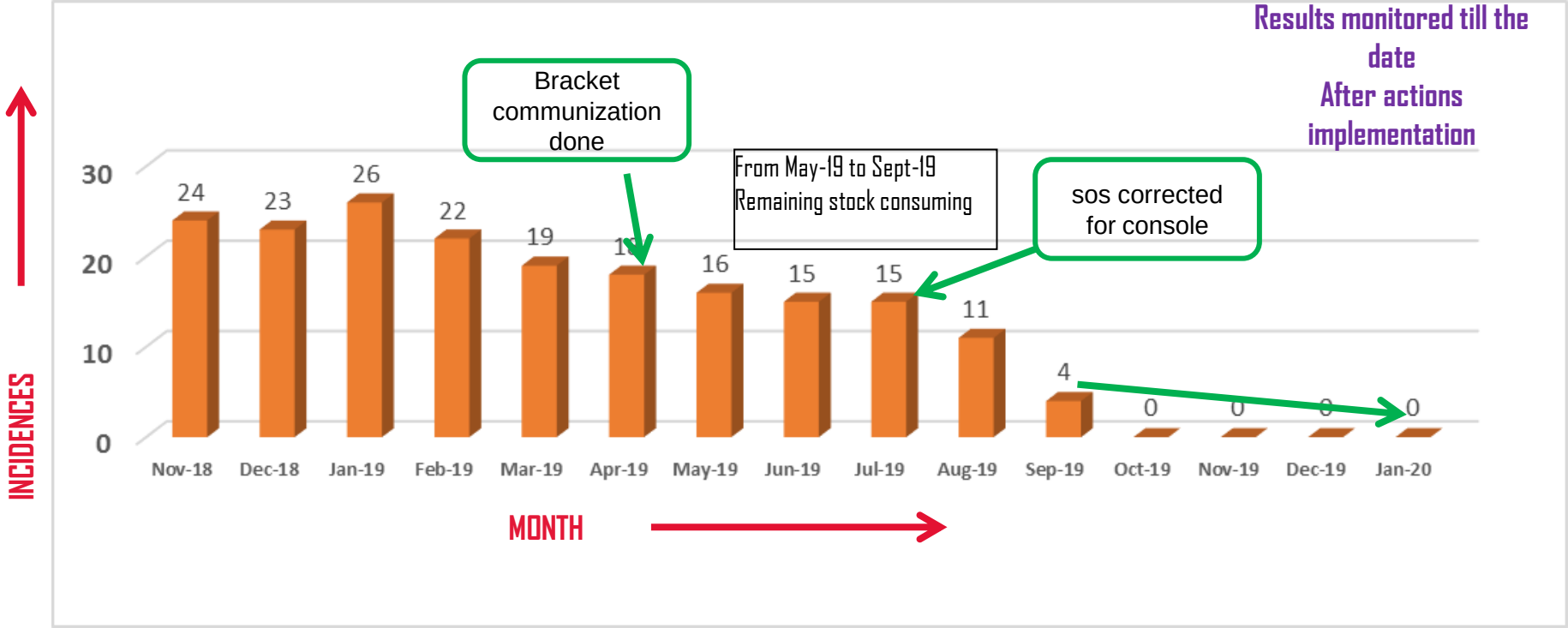
6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
3	No significant difference in parts Bracket wrong fitted.	Bracket commonization done .	25.04.2019	S jadhav	Closed

Bracket communization done for all Scorpio model

7. Check & confirm the results & effects

7.1 Verify Results:



Inference – After implementing bracket no gap observe between console and IP at final roll down and downstream

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:-

S N	Action Taken	Actual date	Responsibility	Status
1	Bracket communization done for all Scorpio model	2.7.19	Amit Tile	Closed
2	EN Copy	25.04.19	Santosh Jadhav	Closed.
3	SOS Updated	2.7.19	Amit Tile	Closed.
4	PFMEA for Console to IP gap	25.4.19	Suhas Aher	Closed.
5	Drawing updated	2.7.19	Santosh Jadhav	Closed.
6	Process Audit check point added.	25.09.19	Amit Tile	Closed.
7	DRF for IP to console Gap	29.6.19	Amit Tile	Closed.

Check & confirm the results & effects (Cont.....)

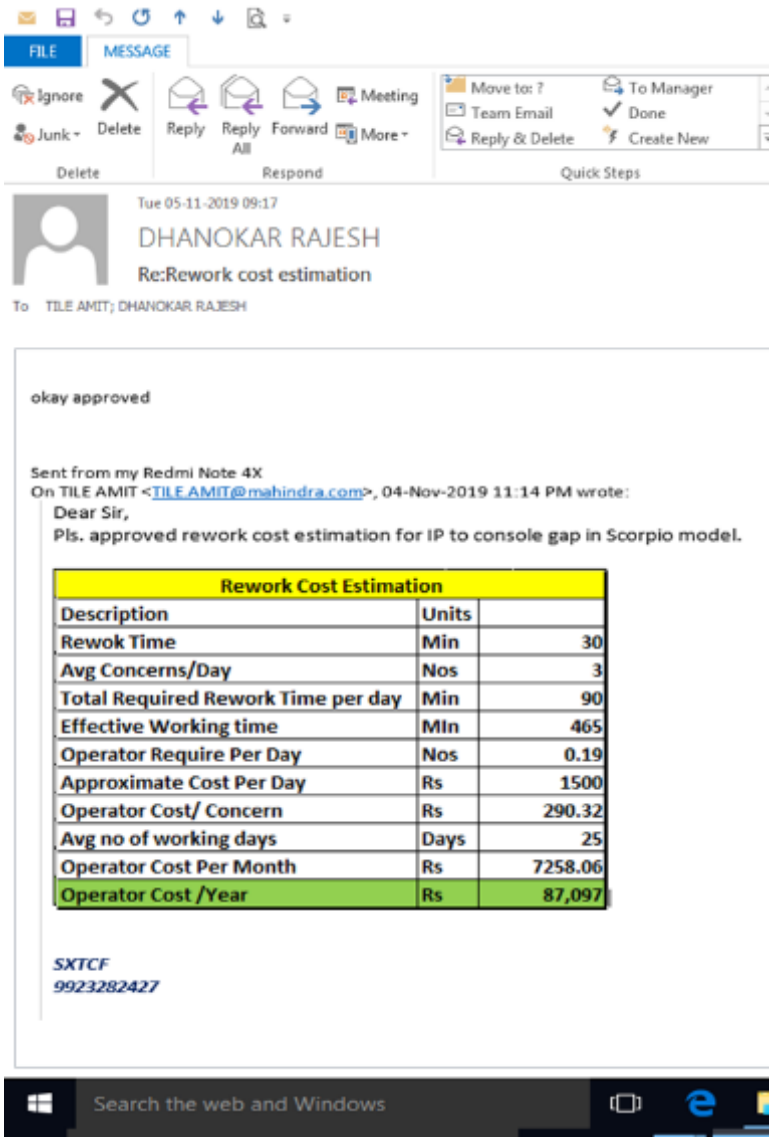
7.3 Tangible & Intangible benefits:

Tangible Benefits -

1)Cost of poor quality saved per year is Rs.87097 per year
2) FRC improved.
3) Scrap cost saved by Rs.18,466/year

In tangible Benefits :

1) Learning of CFT working approach
2) Gemba Culture is learned
3) Associate Morale Increased
4) Improved Customer Satisfaction.
5) Less rework fatigue



Rework Cost Estimation		
Description	Units	
Rewok Time	Min	30
Avg Concerns/Day	Nos	3
Total Required Rework Time per day	Min	90
Effective Working time	MIn	465
Operator Require Per Day	Nos	0.19
Approximate Cost Per Day	Rs	1500
Operator Cost/ Concern	Rs	290.32
Avg no of working days	Days	25
Operator Cost Per Month	Rs	7258.06
Operator Cost /Year	Rs	87,097

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Action Taken	Actual date	Responsibility	Status
1	EN Copy	25.04.19	Santosh Jadhav	Closed.
2	SOS Updated	2.7.19	Amit Tile	Closed.
3	PFMEA for Console to IP gap	25.4.19	Suhas Aher	Closed.
4	Drawing updated	2.7.19	Santosh Jadhav	Closed.
5	Process Audit check point added.	25.9.19	Amit Tile	Closed.
6	DRF for IP to console Gap	29.6.19	Amit Tile	Closed.

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	EN-W110-EN-A003-000123	25.04.19	Santosh Jadhav	Closed.

Mahindra & Mahindra Ltd
Product Development

"Confidential Report"

Page No 1

Mahindra Rise

Engineering Notice Release: W110-EN-A003-000123
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date : 29-Nov-2018
Initiated Date : 25-Feb-2019

Engineering Notice No : W110-EN-A003-000123

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Reference Change Req No : DCR042242

Location : AD-NASIK

Implementation Location : M&M-Nasik

Engineering Notice Description : STANDARDIZATION OF FLOOR CONSOLE MOUNTING BRACKET

Reason : STANDARDIZATION

Quantity : -(Regular)

Reason Code : Quality Improvement

Priority : Mandatory

Note : STANDARDIZATION

Affected Models : W110 ALL

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Released

Owner Name : Sapkal Pravinkumar (301220)

Engineer : Sapkal Pravinkumar (301220)

Approved By : 209584-SONDKAR MAHENDRA

Authorized By : VE-SCORPIO-HEAD_207094-GAUDA MANOJ

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent

Existing : A Part/Assy that is already being used is carried forward as it is to some other parent

Deleted : A Part/Assy that is being used is removed from a Parent

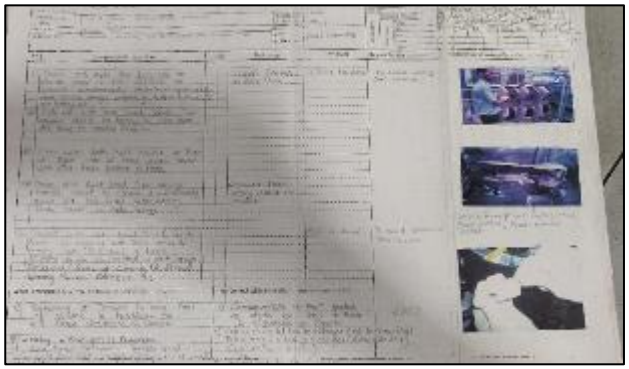
Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

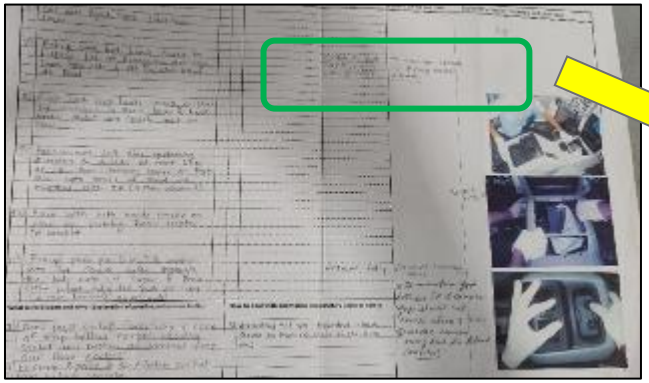
ICA :
O - New part number can be used in place of old part no. & vice versa;
I - New part number can be used in place of old part no. but not vice versa;
Z - Old part number can be used in place of new part no. but not vice versa;
X - No ICA between new and old part numbers;
Y - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
NA - Not Applicable;

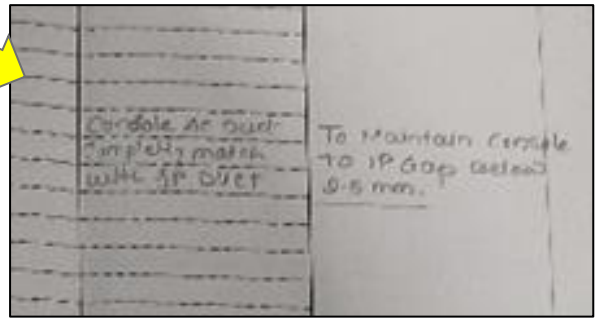
8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
2	SOS Updated	02.07.19	Amit Tile	Closed.





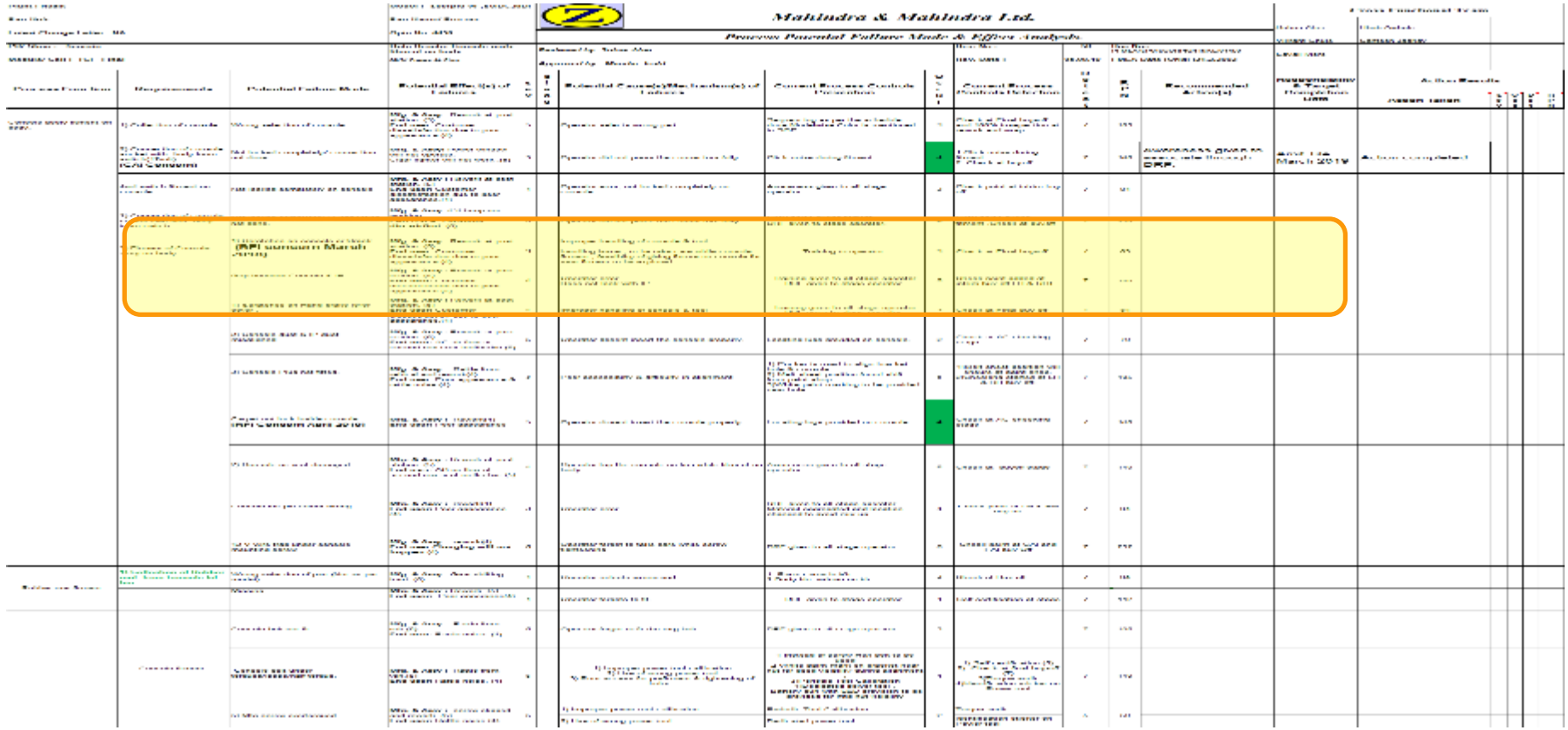


- Associate respect the main step for gear leaver assembly.
- SOS modified for adequate tightening sequence mentioned

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

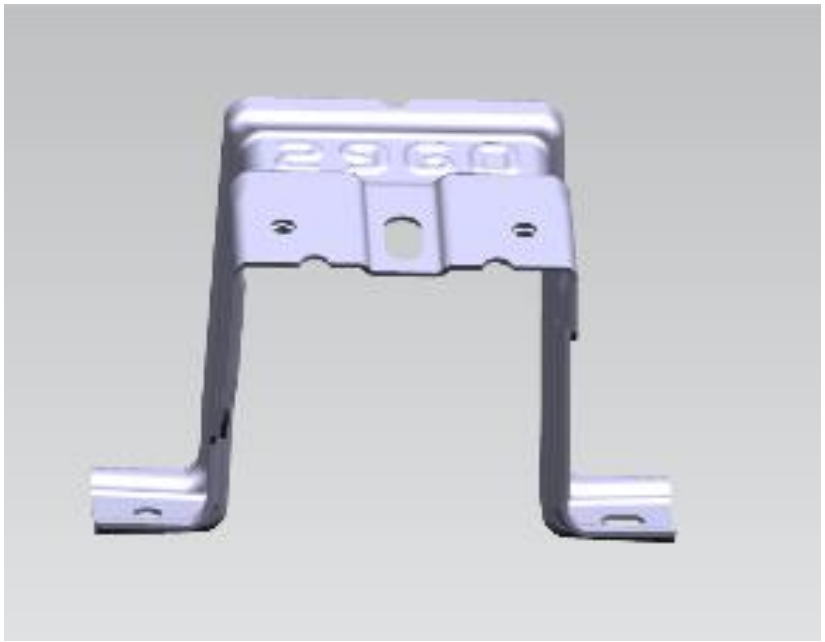
SN	Document	Actual date	Responsibility	Status
3	PFMEA for IP to console gap	25.4.19	Suhas Aher	Closed.



8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
4	Drawing update Console bracket	25.4.19	Santosh Jadhav	Closed.



Bkt. Used for S3 Model Commonized for all Scorpio Models

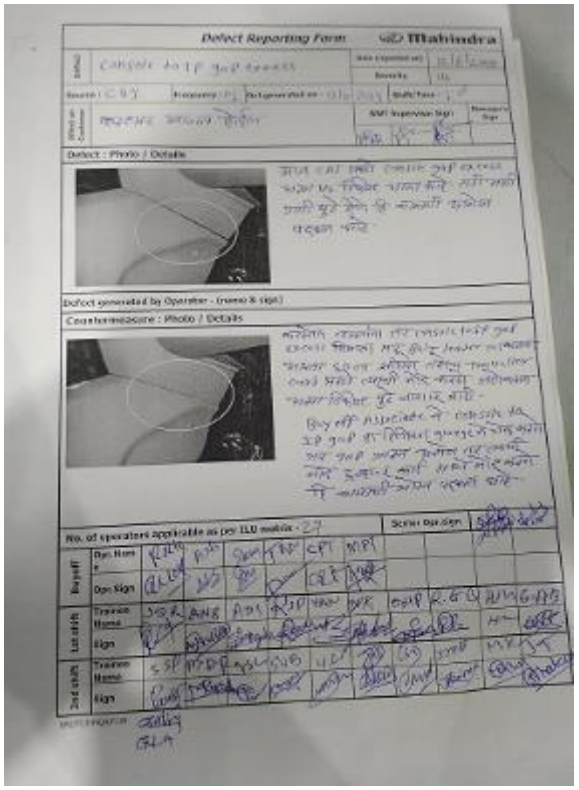
- **Height difference between two bracket was around 8.44mm.**
- **So S3 model bkt was communize for all Scorpio model**

8. Standardization

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
5	DRF to stage associate for IP to console gap	29.06.19	Amit Tile	Closed

DRF

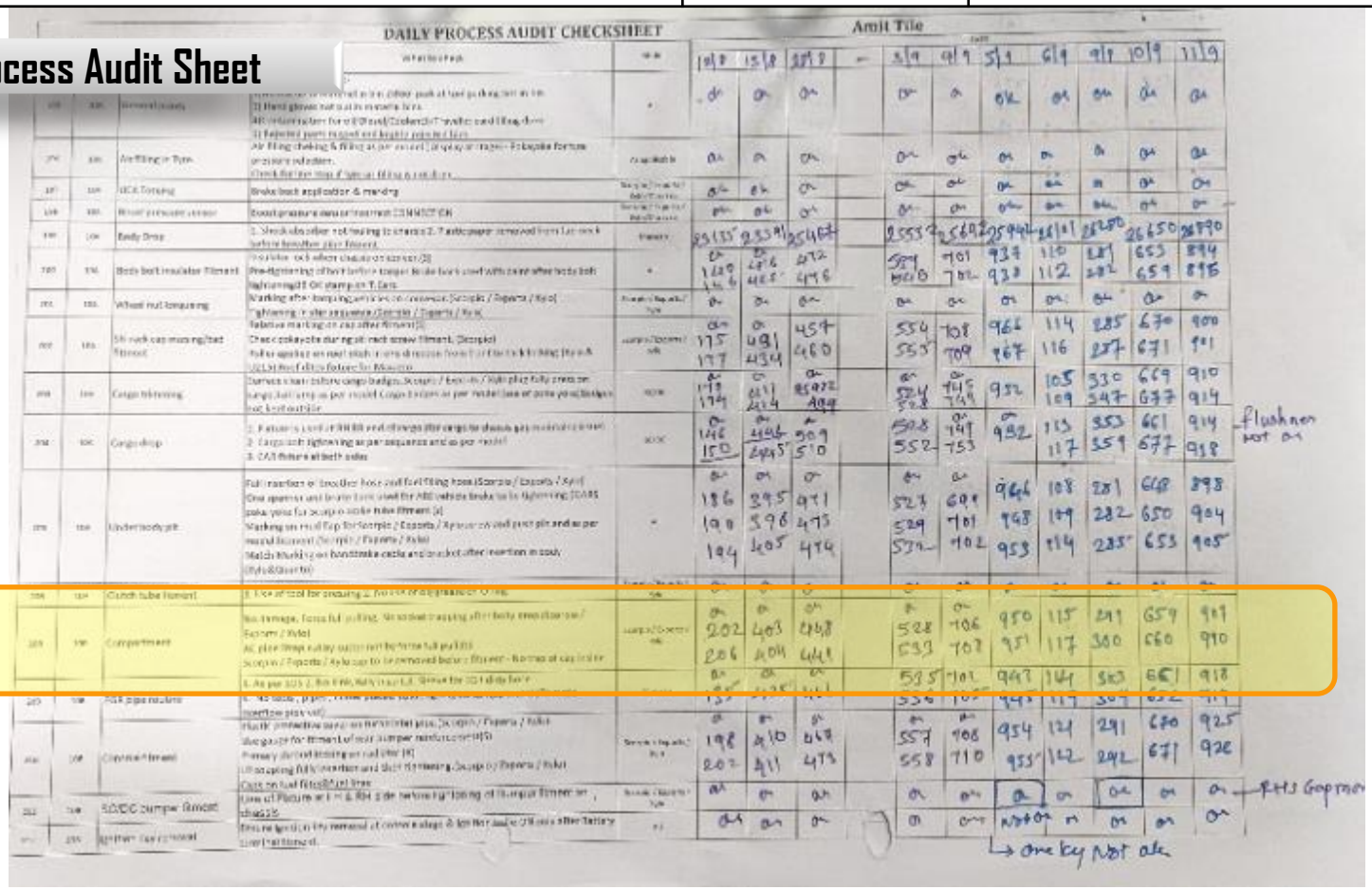


8. Standardization

8.3 Additional Audit & check

6	BWT leaders supervisory check sheet updated.	25.9.19	Amit Tile	Closed.

Process Audit Sheet



9. Summarize & Plan future actions

9.1 Conclusion:

Achieved zero defect in Scorpio model for No gap between console and IP .

9.2 Horizontal Deployment:

All Scorpio models having similar design.

9.3 Lessons Learnt:

Things Gone Wrong :

- Initially only mfg process was suspected as cause of concern.

Things Gone Right :

- Use of associate feedback of CD helped in analysis of defect.
- Process walk though from Press to TCF helped in resolution of defect.

Thank you

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